Sasmita Harini S

sasmitaharini@gmail.com | (+91)-9901245156 | LinkedIn | GitHub

Current Role

Research Fellow, Microsoft India, PROSE Team Aug 2026–Present
Manager: Dr. Vu Le

Education

Indian Institute of Science (IISc), Bengaluru 2022–2026
B.Tech in Mathematics and Computing

Research Interests

Differential Privacy, Privacy-Preserving Machine Learning, Statistical Learning, Large Language Models, Federated Learning, NLP, Optimization, Game Theory

Publications

PROBES: A Bernstein Simplex Approach to Differentially Private OLS Estimation
Under review at Privacy Enhancing Technologies Symposium (PETS 2027)

One Release for All: Private Simple Linear Regression
Under review at InfPriv (NeurIPS Workshop 2026)

Refined Differentially Private Simple Linear Regression via Extension of a Free Lunch Result
SeQureDB Workshop, ACM SIGMOD 2026; Theory and Practice of Differential Privacy (TPDP) 2026

DualEncDecoder: Federated Short-Term Load Forecasting under Heterogeneous Data
FLCA Workshop, AAAI 2026

Refuting LLM-generated Code with Reactive Task Comprehension
ACM ITiCSE 2025

Generating Refute Questions for Concrete, Cluttered Specifications
ACM ICER 2024

Research Internships

Differentially Private Optimization Ongoing
Advised by Prof. Lalitha Sankar and Prof. Oliver Kosut, Arizona State University

- Designed and implemented a differentially private optimizer combining coordinate-wise adaptive clipping (AdaClip) with Adam, featuring an exact time-varying noise-corrected variance tracker and merged EMA recursion for efficient per-sample gradient variance estimation

Privacy-Preserving Language Model Fine-tuning 2025
Advised by Prof. Sayak Ray, IIT Kanpur.

- Evaluated PDP-SGD versus standard DP-SGD for private fine-tuning of Qwen2.5-0.5B on MMLU-Pro and RewardBench-2 using LoRA, studying noise levels and public-dataset size.
- Investigated whether an English public dataset can support private training on multilingual data through cross-lingual transfer.

Evaluation of Sarvam-AI 2025

Advised by Prof. Rahul De, IIM Bangalore.

- Conducted prompt-level jailbreaking experiments to evaluate robustness and safety under adversarial and sensitive prompts.
- Benchmarked medical-diagnosis performance on HealthBench against DeepSeek, Gemma, LLaMA, and Qwen models.

Interactive Audio Chatbot for Mental Health Assessment 2024

Advised by Prof. Deepak Subramanian, IISc

- Built a web application simulating psychologist–patient interactions using ARS-Whisper, SER-Wav2Vec 2.0, and LangChain.
- Fine-tuned LLaMA 3.2 with LoRA and federated learning; used XGBoost for stress-level prediction and RAG for report generation.

Industry Internship

Data Science Intern, Flipkart Jan 2026–July 2026

- Personalized Flipkart's AutoSuggest L2 Ranker by integrating 12 gender and premiumness signals into XGBoost ranking models for AS CTR and Search CTR.
- Worked on the customer support chatbot for the new food delivery app. Designed and build the entire chatbot and worked on enabling follow-up options and two way chat.
- Improved the SOP Writer tool and made it available to all the domains to enable easy SOP writing which usually consumes lot of time and stress.

Privacy Research Intern, PrivaSapien Summer 2025

- Implemented and evaluated differential privacy algorithms for text and enhanced CusText with POS tagging to improve textual DP performance.
- Integrated Tumult Analytics through microservices and automated SQL-to-JSON conversion for privacy-preserving data pipelines.

Research Intern, Indian Institute of Science × TCS Research Summer 2024

- Studied verification-tool prediction using prompt engineering with ChatGPT-4 to predict the suitability of CBMC for validating programs against specified properties.

Awards and Scholarships

- **1st Place**, AMD AI Sprint Hackathon, Bengaluru.
- **AIR 425**, JEE Advanced; **AIR 993**, JEE Main; **69th**, Karnataka CET.
- **NTSE Scholar**, NCERT.
- **Kotak Scholar**, awarded as the highest JEE Advanced ranking female student in the Mathematics and Computing program at IISc.

Technical Skills

Programming: Python, SQL, LaTeX, Bash

ML/AI: PyTorch, Scikit-learn, XGBoost, NLP, LLMs, RAG, LoRA, Deep Learning, GenAI, Prompt Engineering

Relevant Coursework

Applied Linear Algebra and Optimization; Introduction to NLP; Detection and Estimation Theory; Reinforcement Learning; Game Theory; Applied Data Science and AI; Introduction to Scalable Systems; Stochastic Models and Applications; Introduction to AI and ML; Data Structures and Algorithms; Probability and Statistics; Introduction to Basic Analysis; Automata and Computability; Discrete Mathematics.